MIMIC-IV Dataset
QA Benchmark for Coverage

Diversity in RAG Retrieval

Outline

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Why Build a Benchmark on medical data?

- Need to show coverage wins when candidate pools have multi-cluster structure
- Medical data is multi-faceted
  - Aspects: diagnosis, treatment, demographics, comorbidities
- Example of queries:
  - *"How is pneumonia managed in elderly patients with CKD (Complex Kidney Disease)?"*
  - *"What are the symptoms of stroke, and how does the treatment differ if the patient has a prior history of myocardial infarction?"*

Ideal Dataset for Diversity

Given a query, the ideal document pool candidates should have:

- **Redundant central cluster:** dominant topic, many similar documents
- **Complementary cluster(s):** other **relevant** aspects for the query
- **Niche cluster:** smaller set of **relevant** data points
- **Outliers:** noise, or genuine isolated points relevant to the query

Coverage: Facility-Location Objective

- Objective function

$$f(S) = \lambda \sum_{i \in S} utility(i) + \ (\mathbf{1} - \boldsymbol{\lambda}) \sum_{\boldsymbol{i} \in \boldsymbol{D}} \max_{\boldsymbol{j} \in \boldsymbol{S}} \boldsymbol{sim}(\boldsymbol{i}, \boldsymbol{j})$$

  - Every item in the candidate pool D has a representative in S
  - Respects data distribution rather than maximizing spread between items

- Theoretical guarantees
  - g(S) is monotone submodular
  - Greedy selection gives (1 − 1/e) ≈ 0.632 approximation w.r.t. optimal solution
  - No such guarantee exists for *dispersion* (NP-hard)

MIMIC-IV Dataset Overview

- *MIMIC-IV*: tabular clinical data (diagnoses, procedures, meds, labs, measurements)
- *MIMIC-IV-Note*: free-text data
  - ***Discharge Summaries*** *(~330K)*: in-depth notes which overview a patient's history and course throughout a hospitalization
  - *Radiology Reports (~2.3M): reports spanning x-ray, CT, MRI, and ultrasound*

- Discharge summaries allow for design of multi-aspect queries requiring synthesis across patients
- Structured info in MIMIC-IV used for document pool construction and patient usable data

- De-identification: all PHI removed/replaced with ___ placeholders

Ideal Dataset for Coverage: MIMIC-IV?

Cluster Type	Possible MIMIC-IV candidates
Redundant central cluster	Common conditions → hundreds of similar notes
Complementary cluster(s)	Comorbidity combos (pneumonia + CKD + diabetes) → distinct management sub-groups
Niche cluster	Rare complications (e.g., hemorrhagic stroke transformation) → few notes, unique reasoning
Outliers	ICD mismatches, boilerplate copies, de-id artifacts

MIMIC-IV-Note Discharge Summaries

Section	Content	QA Value
Brief Hospital Course	**Clinical reasoning per problem**	**High**
History of Present Illness	Symptom narrative, temporal progression	High
Pertinent Results	Labs, imaging, micro interpretations	High
Discharge Diagnosis	Diagnostic labels (linkable to ICD)	High
Chief Complaint	1–2 lines, reason for admission	High (as embedding metadata)
Past Medical History	Comorbidities list	Medium
- Surgical / Invasive Procedure - Social / Family History - Discharge Medications - Physical Exams	Other minor pieces of information	Low

MIMIC-IV Structured Data

hosp module

Table	Content	Our use
diagnoses_icd	ICD-9/10 codes **per admission**, with `seq_num` priority	ICD labels + query scaffolding
d_icd_diagnoses	ICD code definitions (text descriptions)	Condition names for query templates
patients + admissions	Gender, age, date of death, admission type, insurance, race, discharge location	Patient data for additional context

Discharge summaries: what data to retain?

- **Brief Hospital Course** has highest QA value, but only for complex medicine admissions with # sub-problem lists
    - Complex cases: management reasoning, drug choices with justifications, differentials
- Surgical/simple cases: formulaic procedural timelines, nearly content-free
    - These are noise in any candidate pool: no clinical reasoning

- **History of Present Illnesses** complements BHC: presentation context, symptom timeline, risk factors that BHC presupposes
- **Pertinent Results**: mostly raw lab noise, but imaging interpretations can be high-value

Experimental Pipeline

corpus construction → embeddings → query generation → evaluation

Phase 1: Corpus Construction

1. **Select conditions**
   Query `diagnoses_icd` for ICD groups with ≥ N admissions
2. **Parse and chunk documents**
   Split Discharge Summaries by section header
3. **Attach metadata**
   Per chunk: section name, all ICD codes, demographics, admission type and details
4. **Deduplicate**

→

Output

Per-condition candidate pool D

Phase 2: Embeddings Generation

- Raw chunks are too patient-specific for condition-level queries
- **Add context before chunks to capture full semantics**
- MIMIC metadata is structured → deterministic template to build context

Chunk example

```
Elderly male, White, admitted for Sepsis (SOFA 14).
Chief complaint: fever, hypotension.
Comorbidities: CHF, CKD, COPD.
Section: Brief Hospital Course
<section_content>
```

- Alternative: **whole-document embeddings** via long-context embedding models, depending on which kind of queries we want to construct
  - *"find patients similar to this case"* rather than *"what are the symptoms of X and how do they differ with Y"*

Phase 3: Query Generation via LLM

- Pick primary condition + 1–2 **modifier axes** to ensure multi-aspect
  - Comorbidity: co-occurring ICD codes (e.g. stroke + diabetes)
  - Demographic: age group, gender, race, etc.
  - Complication: post-surgical, recurrent, or secondary diagnoses
- LLM prompt: generate clinical management question requiring modifier knowledge
- **Pre-filter**: only keep queries where *coverage ≠ top-k*
  - assess overall suitability of coverage approaches for this dataset with this embedding strategy
  - $Jaccard(S_{cov}, S_{top-k}) \approx 1$ → skip
  - $g(S_{cov}) - g(S_{top-k}) < threshold$ →skip
- **Gold answer generation**: LLM cites source chunks, each tagged with facet label

Phase 4: Evaluation

Metric	Definition	Measures
Aspect Recall	$AR(S) = \frac{\lvert\{f \in F : \exists\, doc \in S \text{ } with\ facet\ f\}\rvert}{\lvert F \rvert}$	Fraction of annotated facets covered by selected k chunks
Fac-Loc Score	$g(S) = \sum_{i \in D} \max_{j \in S} sim(i, j)$	Representativeness of S over candidate pool D
Jaccard	$J(S_{cov}, S_{top-k})$	Overlap with baseline. J≈1: no change
DocRec / FactRec	Gold doc/fact fraction recovered	How many gold documents/facts are in the selection

Methods: top-k (baseline), MMR, gMMR, facility-location

Pros & Cons

Pros

- Multi-aspect queries tailored to coverage hypothesis
- Dataset seems suitable for coverage
- Gold sets with facet labels enable granular evaluation metrics

Cons

- Discharge notes may be complex to parse and interpret
- De-identification detrimental for semantics
- No guarantee about embedding distribution with the contextual prefix
- Query is still LLM-generated
- LLM gold annotation is expensive and noisy
- Facet labels depend on LLM judgment
- Local inference (DUA). Slower, smaller models
- Many moving parts before first result
- No existing benchmark to compare against
- human validation?

End